Chapter 3
Time Series Applications

3.1. Introduction

This chapter uses tools from chapter 2 to present classic applications. Section 3.2 uses Kalman filter to solve a classic reverse engineering problem that John F. Muth (1960) posed for the "adaptive expectations" model of Milton Friedman. Muth used methods other than the Kalman filter to attack the problem. It turns out that the Kalman filter allows a more concise formulation. Section 3.2 describes how the Kalman filter makes an appearance in a classic paper of Jovanovic (1979). Section 3.3 puts several of our chapter 2 tools to study properties of a workhorse benchmark model in macroeconomics and public finance, the linear-quadratic model of consumption smoothing. We use the model as a vehicle to introduce the complete and incomplete markets models that we shall encounter often in later chapters. Section 3.5 uses our chapter 2 to present Robert E. Lucas's (1976) Critique of some *circa* 1970's econometric policy evaluation procedures, while section 3.6 describes responses to Lucas's Critique.

3.2. Applications of the Kalman filter

3.2.1. Muth's reverse engineering exercise

Phillip Cagan (1956) and Milton Friedman (1957) posited that to form expectations of future values of a scalar y_t, people use the following "adaptive expectations" scheme:

$$y^*_{t+1} = K \sum_{j=0}^{\infty} (1-K)^j \, y_{t-j} \tag{3.2.1a}$$

or

$$y^*_{t+1} = (1-K)\, y^*_t + K y_t, \tag{3.2.1b}$$

where $K \in (0,1)$ and where y^*_{t+1} is people's expectation of y at some future date, perhaps $t+1$.[1] Friedman used this scheme to describe people's forecasts of future income. Cagan used it to model people's forecasts of inflation during hyperinflations. Cagan and Friedman did not assert that the scheme is an optimal one, and did not otherwise fully defend it. Muth (1960) wanted to understand the circumstances under which such a forecasting scheme would be optimal for some horizon. Therefore, he sought a stochastic process for y_t such that equation (3.2.1) would be the formula for a conditional expectation of y_{t+j} conditioned on $y_t, y_{t-1}, \ldots$ for some integer horizon $j \geq 1$. In effect, he posed and solved an "inverse optimal prediction" problem of the form "You give me the forecasting scheme; I have to find a stochastic process that makes the scheme optimal." Muth solved the problem using classical (nonrecursive) methods. The Kalman filter, which first appeared in print in the same year as Muth's paper (Kalman, 1960), lets us solve Muth's inverse optimal prediction problem quickly.

Muth studied the model

$$x_{t+1} = x_t + w_{t+1} \tag{3.2.2a}$$

$$y_t = x_t + v_t, \tag{3.2.2b}$$

[1] See Hamilton (1994) and Kim and Nelson (1999) for diverse applications of the Kalman filter. Appendix B (see Technical Appendixes) briefly describes a discrete-state nonlinear filtering problem.

where y_t, x_t are scalar random processes, and w_{t+1}, v_t are mutually independent i.i.d. Gaussian random processes with means of zero and variances $Ew_{t+1}^2 = Q, Ev_t^2 = R$, and $Ev_s w_{t+1} = 0$ for all t, s. The initial condition is that x_0 is Gaussian with mean $\hat{x}_0$ and variance Σ_0. Muth sought formulas for $\hat{x}_{t+1} = E[x_{t+1}|y^t]$, where $y^t = [y_t, \ldots, y_0]$.

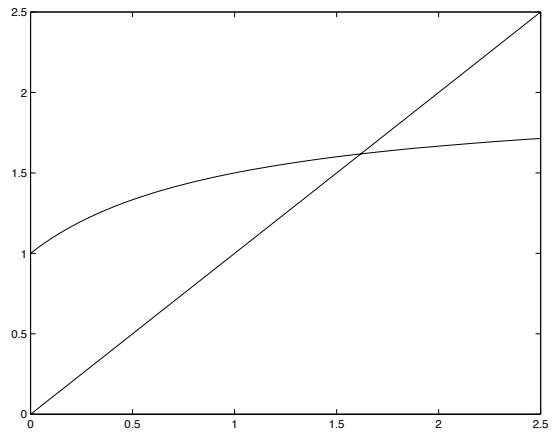

Figure 3.2.1: Graph of $f(\Sigma) = \frac{\Sigma(R+Q)+QR}{\Sigma+R}$, $Q = R = 1$, against the 45-degree line. Iterations on the Riccati equation for Σ_t converge to the fixed point.

For this problem, $A = 1, CC' = Q, G = 1$, making the Kalman filtering equations become

$$K_t = \frac{\Sigma_t}{\Sigma_t + R} \tag{3.2.3a}$$

$$\Sigma_{t+1} = \Sigma_t + Q - \frac{\Sigma_t^2}{\Sigma_t + R}. \tag{3.2.3b}$$

The second equation can be rewritten

$$\Sigma_{t+1} = \frac{\Sigma_t (R+Q) + QR}{\Sigma_t + R}. \tag{3.2.4}$$

For $Q = R = 1$, the coordinate axis of Figure 3.2.1 plots the function $f(\Sigma) = \frac{\Sigma(R+Q)+QR}{\Sigma+R}$ appearing on the right side of equation (3.2.4) for values $\Sigma \geq 0$

against the 45-degree line. Note that $f(0) = Q$. This graph identifies the fixed point of iterations on $f(\Sigma)$ as the intersection of $f(\cdot)$ and the 45-degree line. That the slope of $f(\cdot)$ is less than unity at the intersection assures us that the iterations on f will converge as $t \to +\infty$ starting from any $\Sigma_0 \geq 0$.[2]

Muth studied the solution of this problem as $t \to \infty$. Evidently, $\Sigma_t \to \Sigma_\infty \equiv \Sigma$ is the fixed point of a graph like Figure 3.2.1. Then $K_t \to K$ and the formula for $\hat{x}_{t+1}$ becomes

$$\hat{x}_{t+1} = (1 - K)\,\hat{x}_t + K y_t \tag{3.2.5}$$

where $K = \frac{\Sigma}{\Sigma + R} \in (0, 1)$. This is a version of Cagan's and Friedman's adaptive expectations formula. It can be shown that $K \in [0, 1]$ is an increasing function of $\frac{Q}{R}$. Thus, K is the fraction of the innovation a_t that should be regarded as 'permanent' and $1 - K$ is the fraction that is purely transitory. Iterating backward on equation (3.2.5) gives $\hat{x}_{t+1} = K \sum_{j=0}^{t} (1-K)^j y_{t-j} + (1-K)^{t+1} \hat{x}_0$, which is a version of Cagan and Friedman's geometric distributed lag formula. Using equations (3.2.2), we find that $E[y_{t+j}|y^t] = E[x_{t+j}|y^t] = \hat{x}_{t+1}$ for all $j \geq 1$. This result in conjunction with equation (3.2.5) establishes that the adaptive expectation formula (3.2.5) gives the optimal forecast of y_{t+j} for all horizons $j \geq 1$. This finding is remarkable because for most processes, the optimal forecast will depend on the horizon. That there is a single optimal forecast for all horizons justifies the term *permanent income* that Milton Friedman (1955) chose to describe the forecast of income.

The dependence of the forecast on horizon can be studied using the formulas

$$E\left[x_{t+j}|y^{t-1}\right] = A^j \hat{x}_t \tag{3.2.6a}$$

$$E\left[y_{t+j}|y^{t-1}\right] = G A^j \hat{x}_t \tag{3.2.6b}$$

In the case of Muth's example,

$$E\left[y_{t+j}|y^{t-1}\right] = \hat{y}_t = \hat{x}_t \quad \forall j \geq 0.$$

For Muth's model, the innovations representation is

$$\begin{aligned} \hat{x}_{t+1} &= \hat{x}_t + K a_t \\ y_t &= \hat{x}_t + a_t, \end{aligned} \tag{3.2.7}$$

[2] For the values $Q = R = 1$ used in Figure 3.2.1, it can be verified that the positive solution of $\Sigma = 1.6180339$, that $K = .6180339$, and that the negative solution of the algebraic Ricatti equation is $\Sigma = -.6180339$. The golden ratio equals $.6180339$, the unique positive real number whose reciprocal equals 1 plus itself.

where $a_t = y_t - E[y_t|y_{t-1}, y_{t-2}, \ldots]$. The innovations representation implies that

$$y_{t+1} - y_t = a_{t+1} + (K-1)\, a_t. \tag{3.2.8}$$

Equation (3.2.8) represents $\{y_t\}$ as a process whose first difference is a first-order moving average process. Notice how Friedman's adaptive expectations coefficient K appears in this representation.

3.2.2. Jovanovic's application

In chapter 7, we will describe a version of Jovanovic's (1979) matching model, at the core of which is a "signal-extraction" problem that simplifies Muth's problem. Let x_t, y_t be scalars with $A = 1, C = 0, G = 1, R > 0$. Let x_0 be Gaussian with mean μ and variance Σ_0. Interpret x_t (which is evidently constant with this specification) as the hidden value of θ, a "match parameter". Let y^t denote the history of y_s from $s = 0$ to $s = t$. Define $m_t \equiv \hat{x}_{t+1} \equiv E[\theta|y^t]$ and $\Sigma_{t+1} = E(\theta - m_t)^2$. Then the Kalman filter becomes

$$m_t = (1 - K_t)\, m_{t-1} + K_t y_t \tag{3.2.9a}$$

$$K_t = \frac{\Sigma_t}{\Sigma_t + R} \tag{3.2.9b}$$

$$\Sigma_{t+1} = \frac{\Sigma_t R}{\Sigma_t + R}. \tag{3.2.9c}$$

The recursions are to be initiated from (m_{-1}, Σ_0), a pair that embodies all "prior" knowledge about the position of the system. It is easy to see from Figure 3.2.1 that when $CC' \equiv Q = 0$, $\Sigma = 0$ is the limit point of iterations on equation (3.2.9c) starting from any $\Sigma_0 \geq 0$. Thus, the value of the match parameter is eventually learned.

It is instructive to write equation (3.2.9c) as

$$\frac{1}{\Sigma_{t+1}} = \frac{1}{\Sigma_t} + \frac{1}{R}. \tag{3.2.10}$$

The reciprocal of the variance is often called the *precision* of the estimate. According to equation (3.2.10) the precision increases without bound as t grows, and $\Sigma_{t+1} \to 0$.[3]

[3] As a further special case, consider when there is zero precision initially ($\Sigma_0 = +\infty$). Then solving the difference equation (3.2.10) gives $\frac{1}{\Sigma_t} = t/R$. Substituting this into equations

We can represent the Kalman filter in the form

$$m_{t+1} = m_t + K_{t+1} a_{t+1}$$

which implies that

$$E\left(m_{t+1} - m_t\right)^2 = K_{t+1}^2 \sigma_{a,t+1}^2$$

where $a_{t+1} = y_{t+1} - m_t$ and the variance of a_t is equal to $\sigma_{a,t+1}^2 = (\Sigma_{t+1} + R)$ from equation (6.7.5). This implies

$$E\left(m_{t+1} - m_t\right)^2 = \frac{\Sigma_{t+1}^2}{\Sigma_{t+1} + R}.$$

For the purposes of our discrete-time counterpart of the Jovanovic model in chapter 7, it will be convenient to represent the motion of m_{t+1} by means of the equation

$$m_{t+1} = m_t + g_{t+1} u_{t+1}$$

where $g_{t+1} \equiv \left(\frac{\Sigma_{t+1}^2}{\Sigma_{t+1}+R}\right)^{.5}$ and u_{t+1} is i.i.d. normalized and standardized with mean zero and variance 1 constructed to obey $g_{t+1} u_{t+1} \equiv K_{t+1} a_{t+1}$.

3.3. LQ permanent income model

To illustrate several key ideas of this chapter, this section describes a linear quadratic savings problem whose solution is a rational expectations version of the permanent income model of Friedman (1956) and Hall (1978). We use this model as a vehicle for illustrating impulse response functions, alternative notions of the state, the idea of cointegration, and an invariant subspace method.

The LQ permanent income model is a modification (and not quite a special case, for reasons that will be apparent later) of the following incomplete markets "savings problem" to be studied in chapter 18. A consumer has preferences over consumption streams that are ordered by the utility functional

$$E_0 \sum_{t=0}^{\infty} \beta^t u\left(c_t\right) \tag{3.3.1}$$

(3.2.9) gives $K_t = (t+1)^{-1}$, so that the Kalman filter becomes $m_0 = y_0$ and $m_t = [1-(t+1)^{-1}]m_{t-1} + (t+1)^{-1} y_t$, which implies that $m_t = (t+1)^{-1} \sum_{s=0}^{t} y_t$, the sample mean, and $\Sigma_t = R/t$.

where E_t is the mathematical expectation conditioned on the consumer's time t information, c_t is time t consumption, $u(c)$ is a strictly concave one-period utility function, and $\beta \in (0,1)$ is a discount factor. The consumer maximizes (3.3.1) by choosing a consumption, borrowing plan $\{c_t, b_{t+1}\}_{t=0}^{\infty}$ subject to the sequence of budget constraints

$$c_t + b_t = R^{-1} b_{t+1} + y_t \tag{3.3.2}$$

where $\{y_t\}$ is an exogenous stationary endowment process, R is a constant gross risk-free interest rate, b_t is one-period risk-free debt maturing at t, and b_0 is a given initial condition. It is important to note that for $t \geq 1$, b_t is chosen at time $t-1$ and that b_0 is a given initial condition at time $t = 0$. We shall assume that $R^{-1} = \beta$. For example, we might assume that the endowment process has the state-space representation

$$z_{t+1} = \check{A} z_t + \check{C} w_{t+1} \tag{3.3.3a}$$

$$y_t = \check{G} z_t \tag{3.3.3b}$$

where w_{t+1} is an i.i.d. process with mean zero and identity contemporaneous covariance matrix, $\check{A}$ is a stable matrix, its eigenvalues being strictly below unity in modulus, and $\check{G}$ is a selection vector that identifies y with a particular linear combination of the z_t. We impose the following condition on the consumption, borrowing plan:

$$E_0 \sum_{t=0}^{\infty} \beta^t b_t^2 < +\infty. \tag{3.3.4}$$

This condition suffices to rule out Ponzi schemes. The *state* vector confronting the household at t is $[\, b_t \quad z_t \,]'$, where b_t is its one-period debt falling due at the beginning of period t and z_t contains all variables useful for forecasting its future endowment. We impose this condition to rule out an always-borrow scheme that would allow the household to enjoy bliss consumption forever. The purpose of imposing this condition is to make the solution resemble the solution of problems to be studied in chapter 18 that impose nonnegativity on the consumption path. First-order conditions for maximizing (3.3.1) subject to (3.3.2) are[4]

$$E_t u'(c_{t+1}) = u'(c_t), \quad \forall t \geq 0. \tag{3.3.5}$$

[4] We shall study how to derive this first-order condition in detail in later chapters.

For the rest of this section we assume the quadratic utility function $u(c_t) = -.5(c_t - \gamma)^2$, where $\gamma > 0$ is a bliss level of consumption. Then (3.3.5) implies[5]

$$E_t c_{t+1} = c_t. \tag{3.3.6}$$

Along with the quadratic utility specification, we allow consumption c_t to be negative.[6]

To deduce the optimal decision rule, we want to solve the system of difference equations formed by (3.3.2) and (3.3.6) subject to the boundary condition (3.3.4). To accomplish this, solve (3.3.2) forward and impose $\lim_{T\to+\infty} \beta^T b_{T+1} = 0$ to get

$$b_t = \sum_{j=0}^{\infty} \beta^j \left(y_{t+j} - c_{t+j}\right). \tag{3.3.7}$$

Imposing $\lim_{T\to+\infty} \beta^T b_{T+1} = 0$ suffices to impose (3.3.4) on the debt path. Take conditional expectations on both sides of (3.3.7) and use (3.3.6) and the law of iterated expectations to deduce

$$b_t = \sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \frac{1}{1-\beta} c_t \tag{3.3.8}$$

or

$$c_t = (1-\beta)\left[\sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - b_t\right]. \tag{3.3.9}$$

If we define the net rate of interest r by $\beta = \frac{1}{1+r}$, we can also express this equation as

$$c_t = \frac{r}{1+r}\left[\sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - b_t\right]. \tag{3.3.10}$$

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[5] A linear marginal utility is essential for deriving (3.3.6) from (3.3.5). Suppose instead that we had imposed the following more standard assumptions on the utility function: $u'(c) > 0, u''(c) < 0, u'''(c) > 0$ and required that $c \geq 0$. The Euler equation remains (3.3.5). But $u''' > 0$ implies via Jensen's inequality that $E_t u'(c_{t+1}) > u'(E_t c_{t+1})$. This inequality together with (3.3.5) implies that $E_t c_{t+1} > c_t$ (consumption is said to be a 'submartingale'), so that consumption stochastically diverges to $+\infty$. The consumer's savings also diverge to $+\infty$. Chapter 18 discusses such 'precautionary savings' divergence.

[6] That c_t can be negative explains why we impose condition (3.3.4) instead of an upper bound on the level of borrowing, such as the natural borrowing limits of chapters 9, 18, and 19.

Equation (3.3.9) or (3.3.10) asserts that consumption equals economic *income*, namely, a constant marginal propensity to consume or interest factor $\frac{r}{1+r}$ times the sum of nonfinancial wealth $\sum_{j=0}^{\infty} \beta^j E_t y_{t+j}$ and financial wealth $-b_t$. Notice that we can use a version of our formula (2.4.17) in (3.3.9) or (3.3.10) to obtain

$$c_t = (1-\beta)\left[\check{G}\left(I-\beta\check{A}\right)^{-1} z_t - b_t\right], \tag{3.3.11}$$

a "consumption function" that represents c_t as a function of the *state* $[b_t, z_t]$ that confronts the household, where from (3.3.3) z_t contains present information that is useful for forecasting the endowment process.

Pulling together our preceding results, we can regard z_t, b_t as the time t state, where z_t is an *exogenous* component of the state and b_t is an *endogenous* component of the state vector. The system can be represented as

$$\begin{aligned}
z_{t+1} &= \check{A} z_t + \check{C} w_{t+1} \\
b_{t+1} &= b_t + \check{G}\left[\left(I-\beta\check{A}\right)^{-1}\left(\check{A}-I\right)\right] z_t \\
y_t &= \check{G} z_t \\
c_t &= (1-\beta)\left[\check{G}\left(I-\beta\check{A}\right)^{-1} z_t - b_t\right].
\end{aligned}$$

where again, the last equation comes from applying to decision rule (3.3.10) the appropriate version our formula (2.4.17) for forecasting geometric sums of future values of the random variable y_t.

3.3.1. Another representation

Another way to understand the system under the optimal decision rules for b_t, c_t is to recognize that there is a point of view that allows us to regard the state as being c_t together with z_t and to regard b_t as an outcome. Hall (1978) showed that this is a sharp way to summarize the LQ permanent income theory. We proceed to transform the state vector in this way.

To represent an optimal plan for b_t, substitute (3.3.9) into (3.3.2) and after rearranging obtain

$$b_{t+1} = b_t + \left(\beta^{-1}-1\right)\sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \beta^{-1} y_t. \tag{3.3.12}$$

Next, shift (3.3.9) forward one period and eliminate b_{t+1} by using (3.3.2) to obtain

$$c_{t+1} = (1-\beta)\sum_{j=0}^{\infty} E_{t+1}\beta^j y_{t+j+1} - (1-\beta)\left[\beta^{-1}\left(c_t + b_t - y_t\right)\right].$$

If we add and subtract $\beta^{-1}(1-\beta)\sum_{j=0}^{\infty}\beta^j E_t y_{t+j}$ from the right side of the preceding equation and rearrange, we obtain

$$c_{t+1} - c_t = (1-\beta)\sum_{j=0}^{\infty}\beta^j\left(E_{t+1}y_{t+j+1} - E_t y_{t+j+1}\right). \tag{3.3.13}$$

The right side is the time $t+1$ innovation to the expected present value of the endowment process $\{y_t\}$. It is useful to express this innovation in terms of a moving average representation for income y_t. Suppose that the endowment process has the moving average representation[7]

$$y_{t+1} = d\left(L\right)w_{t+1} \tag{3.3.14}$$

where $\{w_{t+1}\}$ is an i.i.d. vector process with $Ew_{t+1} = 0$ and contemporaneous covariance matrix $Ew_{t+1}w'_{t+1} = I$, $d(L) = \sum_{j=0}^{\infty} d_j L^j$, where L is the lag operator, and the household has an information set $w^t = [w_t, w_{t-1}, \ldots,]$ at time t.[8] Then notice that

$$y_{t+j} - E_t y_{t+j} = d_0 w_{t+j} + d_1 w_{t+j-1} + \cdots + d_{j-1}w_{t+1}.$$

It follows that

$$E_{t+1}y_{t+j} - E_t y_{t+j} = d_{j-1}w_{t+1}. \tag{3.3.15}$$

Using (3.3.15) in (3.3.13) gives

$$c_{t+1} - c_t = (1-\beta)\,d\left(\beta\right)w_{t+1}. \tag{3.3.16}$$

The object $d(\beta)$ is the present value of the moving average coefficients in the representation for the endowment process y_t.

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[7] Representation (3.3.3) implies that $d(L) = \check{G}(I - \check{A}L)^{-1}\check{C}$.

[8] A moving average representation for a process y_t is said to be *fundamental* if the linear space spanned by y^t is equal to the linear space spanned by w^t. A time-invariant innovations representation, attained via the Kalman filter as in section 2.7, is by construction fundamental.

After all of this work, we can represent the optimal decision rule for c_t, b_{t+1} in the form of the two equations (3.3.13) and (3.3.8), which we repeat here for convenience:

$$c_{t+1} = c_t + (1-\beta)\sum_{j=0}^{\infty}\beta^j\left(E_{t+1}y_{t+j+1} - E_t y_{t+j+1}\right) \tag{3.3.17}$$

$$b_t = \sum_{j=0}^{\infty}\beta^j E_t y_{t+j} - \frac{1}{1-\beta}c_t. \tag{3.3.18}$$

Equation (3.3.18) asserts that a household's debt due at t equals the expected present value of its endowment minus the expected present value of its consumption stream. A high debt thus indicates a large expected present value of 'surpluses' $y_t - c_t$.

Recalling the form of the endowment process (3.3.3), we can compute

$$E_t\sum_{j=0}^{\infty}\beta^j z_{t+j} = \left(I-\beta\check{A}\right)^{-1} z_t$$

$$E_{t+1}\sum_{j=0}^{\infty}\beta^j z_{t+j+1} = \left(I-\beta\check{A}\right)^{-1} z_{t+1}$$

$$E_t\sum_{j=0}^{\infty}\beta^j z_{t+j+1} = \left(I-\beta\check{A}\right)^{-1}\check{A}z_t.$$

Substituting these formulas into (3.3.17) and (3.3.18) and using (3.3.3a) gives the following representation for the consumer's optimum decision rule:[9]

$$c_{t+1} = c_t + (1-\beta)\check{G}\left(I-\beta\check{A}\right)^{-1}\check{C}w_{t+1} \tag{3.3.19a}$$

$$b_t = \check{G}\left(I-\beta\check{A}\right)^{-1} z_t - \frac{1}{1-\beta}c_t \tag{3.3.19b}$$

$$y_t = \check{G}z_t \tag{3.3.19c}$$

$$z_{t+1} = \check{A}z_t + \check{C}w_{t+1} \tag{3.3.19d}$$

Representation (3.3.19) reveals several things about the optimal decision rule. (1) The *state* consists of the endogenous part c_t and the exogenous part

[9] See appendix A of chapter 18 for a reinterpretation of precisely these outcomes in terms of a competitive equilibrium of a model with a complete set of markets in history- and date-contingent claims to consumption.

z_t.[10] These contain all of the relevant information for forecasting future c, y, b. Notice that financial assets b_t have disappeared as a component of the state because they are properly encoded in c_t. (2) According to (3.3.19), consumption is a random walk with innovation $(1-\beta)d(\beta)w_{t+1}$ as implied also by (3.3.16). This outcome confirms that the Euler equation (3.3.6) is built into the solution. That consumption is a random walk of course implies that it does not possess an asymptotic stationary distribution, at least so long as z_t exhibits perpetual random fluctuations, as it will generally under (3.3.3).[11] This feature is inherited partly from the assumption that $\beta R = 1$. (3) The impulse response function of c_t is a "box": for all $j \geq 1$, the response of c_{t+j} to an increase in the innovation w_{t+1} is $(1-\beta)d(\beta) = (1-\beta)\check{G}(I-\beta\check{A})^{-1}\check{C}$. (4) Solution (3.3.19) reveals that the joint process c_t, b_t possesses the property that Granger and Engle (1987) called *cointegration*. In particular, *both* c_t and b_t are non-stationary because they have unit roots (see representation (3.3.12) for b_t), but there is a linear combination of c_t, b_t that *is* stationary provided that z_t is stationary. From (3.3.18), the linear combination is $(1-\beta)b_t + c_t$. Accordingly, Granger and Engle would call $[\,(1-\beta) \quad 1\,]$ a cointegrating vector that, when applied to the nonstationary vector process $[\,b_t \quad c_t\,]'$, yields a process that is asymptotically stationary. Equation (3.3.8) can be arranged to take the form

$$(1-\beta)\, b_t + c_t = (1-\beta)\, E_t \sum_{j=0}^{\infty} \beta^j y_{t+j}, \tag{3.3.20}$$

which asserts that the 'cointegrating residual' on the left side equals the conditional expectation of the geometric sum of future incomes on the right.[12]

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[10] Something subtle is going on in system (3.3.17)-(3.3.18): the *state* variable c_t is actually responding to the shocks w_t to z_t that shape time t nonfinancial income y_t while the "outcome" variable b_t has actually been chosen at $t-1$ and so it is predetermined. It can be expressed as an exact function of the time t "state" variables c_t, z_t only because c_t adjusts to z_t in just the right way.

[11] The failure of consumption to converge will occur again in chapter 18 when we drop quadratic utility and assume that consumption must be nonnegative.

[12] See Campbell and Shiller (1988) and Lettau and Ludvigson (2001, 2004) for interesting applications of related ideas.

3.3.2. Debt dynamics

If we subtract equation (3.3.19b) evaluated at time t from equation (3.3.19b) evaluated at time $t+1$, we obtain

$$b_{t+1} - b_t = \check{G}\left(I - \beta\check{A}\right)^{-1}(z_{t+1} - z_t) - \frac{1}{1-\beta}(c_{t+1} - c_t).$$

Substituting $z_{t+1} - z_t = (\check{A} - I)z_t + \check{C}w_{t+1}$ and equation (3.3.19a) into the above equation and rearranging gives

$$b_{t+1} - b_t = \check{G}\left(I - \beta\check{A}\right)^{-1}\left(\check{A} - I\right)z_t. \tag{3.3.21}$$

This equation indicates how b_{t+1} is predetermined at time t as a function of b_t and z_t.

3.3.3. Two classic examples

We illustrate formulas (3.3.19) with the following two examples. In both examples, the endowment follows the process $y_t = z_{1t} + z_{2t}$ where

$$\begin{bmatrix} z_{1t+1} \\ z_{2t+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} + \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}\begin{bmatrix} w_{1t+1} \\ w_{2t+1} \end{bmatrix} \tag{3.3.22}$$

where w_{t+1} is an i.i.d. 2×1 process distributed as $\mathcal{N}(0, I)$. Here z_{1t} is a permanent component of y_t while z_{2t} is a purely transitory component.

Example 1. Assume that the consumer observes the state z_t at time t. This implies that the consumer can construct w_{t+1} from observations of z_{t+1} and z_t. Application of formulas (3.3.19) implies that

$$c_{t+1} - c_t = \sigma_1 w_{1t+1} + (1-\beta)\,\sigma_2 w_{2t+1}. \tag{3.3.23}$$

Since $1-\beta = \frac{r}{1+r}$ where $R = (1+r)$, formula (3.3.23) shows how an increment $\sigma_1 w_{1t+1}$ to the permanent component of income z_{1t+1} leads to a permanent one-for-one increase in consumption and no increase in savings $-b_{t+1}$; but how the purely transitory component of income $\sigma_2 w_{2t+1}$ leads to a permanent increment in consumption by a fraction $(1-\beta)$ of transitory income, while the remaining fraction β is saved, leading to a permanent increment in savings $-b_{t+1}$. Application of formula (3.3.21) to this example shows that

$$b_{t+1} - b_t = -z_{2t} = -\sigma_2 w_{2t}, \tag{3.3.24}$$

which confirms that none of $\sigma_1 w_{1t}$ is saved, while all of $\sigma_2 w_{2t}$ is saved.

Example 2. We build on Muth's classic reverse engineering exercise that we described in section 3.2.1. Thus, we assume that the income process $\{y_t\}$ obeys state space system (3.2.2) and that the consumer observes y_t, and its history up to t, but not the hidden state x_t at time t. In terms of the notation used in system (3.3.22), this amounts to assuming that the consumer observes $y_t = z_{1t} + z_{2t}$ but not z_{1t} and z_{2t} separately. Under this assumption, it is appropriate to use an *innovation representation* to form $\check{A}, \check{C}, \check{G}$ in formulas (3.3.19). In particular, using the innovations representation (3.2.7) derived in subsection 3.2.1, the appropriate state-space representation for y_t is

$$\begin{bmatrix} y_{t+1} \\ a_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -(1-K) \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ a_t \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} a_{t+1}$$
$$y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ a_t \end{bmatrix}$$

where K is the Kalman gain and $a_t = y_t - E[y_t | y^{t-1}]$. From subsection 3.2.1, we know that $K \in [0,1]$ and that K increases as $\frac{\sigma_1^2}{\sigma_2^2}$ increases, i.e., as the ratio of the variance of the permanent shock to the variance of the transitory shock to income increases. Applying formulas (3.3.19) implies

$$c_{t+1} - c_t = [1 - \beta\,(1-K)]\, a_{t+1} \tag{3.3.25}$$

where the endowment process can now be represented in terms of the univariate innovation to y_t as

$$y_{t+1} - y_t = a_{t+1} - (1-K)\, a_t. \tag{3.3.26}$$

Equation (3.3.26) indicates that the consumer regards a fraction K of an innovation a_{t+1} to y_{t+1} as *permanent* and a fraction $1-K$ as purely transitory. He permanently increases his consumption by the full amount of his estimate of the permanent part of a_{t+1}, but by only $(1-\beta)$ times his estimate of the purely transitory part of a_{t+1}. Therefore, in total he permanently increments his consumption by a fraction $K + (1-\beta)(1-K) = 1 - \beta(1-K)$ of a_{t+1} and saves the remaining fraction $\beta(1-K)$ of a_{t+1}. According to equation (3.3.26), the first difference of income is a first-order moving average, while (3.3.25) asserts that the first difference of consumption is i.i.d. Application of formula (3.3.21) to this example shows that

$$b_{t+1} - b_t = (K-1)\, a_t, \tag{3.3.27}$$

which indicates how the fraction K of the innovation to y_t that is regarded as permanent influences the fraction of the innovation that is saved.

3.3.4. Spreading consumption cross section

Starting from an arbitrary initial distribution for c_0 and say the asymptotic stationary distribution for z_0, if we were to apply formulas (2.4.9) and (2.4.10) to the state space system (3.3.19), the common unit root affecting c_t, b_t would cause the time t variance of c_t to grow linearly with t. If we think of the initial distribution as describing the joint distribution of c_0, b_0 for a cross section of ex ante identical households 'born at time 0, then these formulas would describe the evolution of the cross-section for b_t, c_t as the population of households ages. The distribution would spread out.[13]

3.3.5. Invariant subspace approach

We can glean additional insights about the structure of the optimal decision rule by solving the decision problem in a mechanical but quite revealing way that easily generalizes to a host of problems, as we shall see later in chapter 6. We can represent the system consisting of the Euler equation (3.3.6), the budget constraint (3.3.2), and the description of the endowment process (3.3.3) as

$$\begin{bmatrix} \beta & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_{t+1} \\ z_{t+1} \\ c_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -\check{G} & 1 \\ 0 & \check{A} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b_t \\ z_t \\ c_t \end{bmatrix} + \begin{bmatrix} 0 \\ \check{C} \\ C_1 \end{bmatrix} w_{t+1} \tag{3.3.28}$$

where C_1 is an undetermined coefficient. Premultiply both sides by the inverse of the matrix on the left and write

$$\begin{bmatrix} b_{t+1} \\ z_{t+1} \\ c_{t+1} \end{bmatrix} = \tilde{A} \begin{bmatrix} b_t \\ z_t \\ c_t \end{bmatrix} + \tilde{C} w_{t+1}. \tag{3.3.29}$$

We want to find solutions of (3.3.29) that satisfy the no-explosion condition (3.3.4). We can do this by using machinery to be introduced in chapter 6. The

[13] See Deaton and Paxton (1994) and Storesletten, Telmer, and Yaron (2004) for evidence that cross section distributions of consumption spread out with age.

key idea is to discover what part of the vector $[\, b_t \quad z_t \quad c_t \,]'$ is truly a *state* from the view of the decision maker, being inherited from the past, and what part is a *costate* or *jump* variable that can adjust at t. For our problem b_t, z_t are truly components of the state, but c_t is free to adjust. The theory determines c_t at t as a function of the true state variables $[b_t, z_t]$. A powerful approach to determining this function is the following so-called invariant subspace method of chapter 6. Obtain the eigenvector decomposition of $\tilde{A}$:

$$\tilde{A} = V \Lambda V^{-1}$$

where Λ is a diagonal matrix consisting of the eigenvalues of $\tilde{A}$ and V is a matrix of the associated eigenvectors. Order the eigenvalues from smallest absolute value to largest and form partition $\Lambda = \begin{bmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{bmatrix}$ where the eigenvalues in Λ_1 are less than unity in modulus and the eigenvalues in Λ_2 all exceed unity in modulus. Let $V^{-1} \equiv \begin{bmatrix} V^{11} & V^{12} \\ V^{21} & V^{22} \end{bmatrix}$ be the corresponding partition of V^{-1}.[14] Then applying formula (6.5.13) of chapter 6 implies that if (3.3.4) is to hold, the jump variable c_t must satisfy

$$c_t = -\left(V^{22}\right)^{-1} V^{21} \begin{bmatrix} b_t \\ z_t \end{bmatrix}. \tag{3.3.30}$$

Formula (3.3.30) gives the unique value of c_t that ensures that (3.3.4) is satisfied, or in other words, that the state remains in the "stabilizing subspace." Notice that the variables on the right side of (3.3.30) conform with those called for by (3.3.10): $-b_t$ is there as a measure of financial wealth, and z_t is there because it includes all variables that are useful for forecasting the future endowments that appear in (3.3.10).

[14] It can be verifies by hand that for matrix $\tilde{A}$ defined in systems (3.3.28) and (3.3.29), the matrix $\Lambda_2 = \beta^{-1}$, a scalar that exceeds unity.

3.4. Consumption smoothing with complete markets

It is instructive to take the LQ permanent income model studied in section 3.3 and, by altering assumptions about *what* is determined *when* in equation (3.3.7), provide the consumer with the opportunity to avoid all fluctuations in consumption by trading state-contingent claims to future consumption. To start, return to equations (3.3.6) and (3.3.7) that we repeat for convenience:

$$E_t c_{t+1} = c_t. \tag{3.4.1}$$

$$b_t = \sum_{j=0}^{\infty} \beta^j \left(y_{t+j} - c_{t+j}\right). \tag{3.4.2}$$

Recall that (3.4.1) is an Euler equation for consumption, i.e., a first-order necessary condition for constrained optimization, and that (3.4.2) is the outcome of solving in the forward direction the first-order linear difference equation that is the consumer's one-period budget constraint using the method described in appendix B of this chapter. As we did in section 3.3, take conditional expectations on both sides of (3.4.2) and use (3.4.1) and the law of iterated expectations to deduce

$$b_t = \sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \frac{1}{1-\beta} c_t. \tag{3.4.3}$$

As in section 3.3, we assume that non-financial income y_t is governed by the state-space representation (3.3.3), namely,

$$\begin{aligned} z_{t+1} &= \check{A} z_t + \check{C} w_{t+1} \\ y_t &= \check{G} z_t, \end{aligned}$$

where w_{t+1} is an i.i.d. process with mean zero and identity contemporaneous covariance matrix, all eigenvalues of $\check{A}$ are strictly less than $\frac{1}{\sqrt{\beta}}$ in magnitude, and $\check{G}$ is a selection vector that describes y as a particular linear combination of z_t. Computing conditional expectations of $y_{t+j}, j \geq 0$ and substituting into (3.4.3), we obtain

$$c_t = (1-\beta)\left[\check{G}\left(I - \beta\check{A}\right)^{-1} z_t - b_t\right]. \tag{3.4.4}$$

We now pause to stare at equation (3.4.4) and think about alternative assumptions about what is predetermined at t and what is free to adjust to

shocks to z_t, i.e., alternative assumptions about what are *states* and what are *outcomes* at t. In section 3.3, we regarded both b_t and z_t as components of the consumer's state at time t, each having been determined before time t, with the scalar $-b_t$ being the consumer's financial assets in the form of risk-free claims to time t consumption that the consumer purchased at time $t-1$ and the vector z_t being information that is useful for predicting future non-financial incomes $\{y_{t+j}\}_{j=0}^{\infty}$. In section 3.3, we interpreted (3.4.4) as the consumer's optimal decision rule for c_t as a function of these state variables, i.e., the consumer's "consumption function".

By making other assumptions about what variables in (3.4.4) are set before time t (they are "predetermined" at time t) and what variables can be adjusted at time t, we can put the consumer into a system of markets that provide him more opportunities to trade. To show this, we begin by constructing another consumption plan that satisfies both conditions (3.4.1) and (3.4.2) as well as a special case of equation (3.4.4) in which

$$c_t = \bar{c} = c_0, \tag{3.4.5}$$

where c_0 is given by equation (3.3.11) for the optimal time 0 consumption in the section 3.3 incomplete markets version of the model. Because it completely smooths consumption and has the same time 0 consumption, this $c_t = \bar{c}$ plan awards higher discounted utility to the consumer than does the random walk consumption plan implied by decision rule (3.4.4), an assertion that we'll soon prove.

To construct a financial plan that supports constant consumption, temporarily take $\bar{c}$ as given, substitute (3.4.5) into (3.4.4) and solve for the following state-dependent debt function $b(z_t, \bar{c})$:

$$b\left(z_t, \bar{c}\right) = \check{G}\left(I - \beta\check{A}\right)^{-1} z_t - \frac{1}{1-\beta}\bar{c}. \tag{3.4.6}$$

To pin down the constant $\bar{c}$, we assume that the consumer enters time 0 owing debt $\bar{b}_0$ and that $\bar{c}$ solves the following version of (3.4.6) at time $t = 0$:

$$\bar{c} = (1-\beta)\left[\check{G}\left(I - \beta\check{A}\right)^{-1} z_0 - \bar{b}_0\right] \tag{3.4.7}$$

so that $b(z_0, \bar{c}) = \bar{b}_0$ also satisfies equation (3.4.6).

In effect, we have backed ourselves into a model in which each period the consumer trades a complete list of one-period state-contingent securities that

pay off amounts next period that depend on next period's information vector z. To make this work, prices of those securities must induce the consumer to want to smooth consumption completely across both states and time by setting $c_t = \bar{c} = c_0$. We can construct prices of state-contingent securities that convince the consumer to do that. These prices will turn out to imply that the price at time t of one unit of consumption for sure at time $t+1$ is β for all $t \geq 0$, just as it is in the section 3.3 model in which the consumer can trade only a risk-free one-period bond.

Our consumer now faces a sequence of one-period budget constraints whose time $t \geq 0$ component is

$$c_t + b_{t-1}(z_t) = \int q(z_{t+1}|z_t)\, b_t(z_{t+1})\, dz_{t+1} + y_t \tag{3.4.8}$$

where $q(z_{t+1}|z_t)$ is a pricing kernel for one-period state contingent claims and $b_t(z_{t+1})$ is a vector of state-contingent debts due in state z_{t+1} that are chosen at time t. We exogenously set the pricing kernel to

$$q(z_{t+1}|z_t) = \beta\phi(z_{t+1}|z_t), \tag{3.4.9}$$

where $\phi(z_{t+1}\ z_t)$ is the probability density of z_{t+1} conditional on z_t. Kernel (3.4.9) discounts state-contingent prices of claims on consumption at date $t+1$ by the product of the discount factor β and conditional probabilities of events at which claims pay off. This special pricing kernel implies that the price of a one-period risk-free bond is

$$\int q(z_{t+1}|z_t)\, dz_{t+1} = \beta \int \phi(z_{t+1}|z_t)\, dz_{t+1} = \beta, \tag{3.4.10}$$

so that the gross interest rate on one-period risk free loans is β^{-1}.[15]

With pricing kernel (3.4.9), the consumer's time t one-period budget constraint (3.4.8) becomes

$$c_t + b_{t-1}(z_t) = \beta E b_t(z_{t+1})|z_t + y_t. \tag{3.4.11}$$

[15] It would be redundant to add a risk-free one-period bond to the state-contingent one period securities that the consumer can trade in our complete markets model. If that redundant asset were to be added, a no-arbitrage profits argument like the one to be applied repeatedly in chapters 9, 14, and 15 would imply formula (3.4.10).

instead of the section 3.3 budget constraint

$$c_t + b_t = \beta b_{t+1} + y_t \tag{3.4.12}$$

in which b_{t+1} is a scalar to be chosen at time t.

Notice that $\bar{c}$ in equation (3.4.7) equals c_0 given by the optimal decision rule (3.3.11) for the section 3.3 incomplete markets version of the LQ consumption smoothing model.[16] From equation (3.3.19a), we know that in the incomplete markets version of the model

$$c_{t+1} = c_t + hw_{t+1}$$

where $h = (1-\beta)\check{G}(I - \beta\check{A})^{-1}\check{C}$ so that

$$\begin{aligned} E_0 c_t &= c_0 \\ E_0 \left(c_t - c_0\right)^2 &= thh'. \end{aligned}$$

Because the consumer dislikes variance in consumption, he prefers the completely smoothed consumption stream that he chooses in the complete markets version of the model to the random walk consumption stream that he chooses in the incomplete markets version of the model.

We shall study more general complete markets models of consumption smoothing in chapter 9 and more general incomplete markets models in chapter 19. In those chapters, interest rates and state-contingent prices are among the objects that are determined within an equilibrium.

[16] It is an incomplete markets model in the sense that the consumer can trade only risk-free one-period securities instead of the complete set of securities present in the model of this section.

3.5. The Lucas Critique

We can use our chapter 2 tools concisely to describe reasoning that led Robert E. Lucas, Jr., (1976) to criticize a class of procedures that were widely used in applied work at the time that he wrote, and that occasionally still are. Assume a jointly stationary Gaussian stochastic process for $\{x_t, y_t, d_t\}_{t=0}^{\infty}$ generated by a state-space system

$$x_{t+1} = Ax_t + Cw_{t+1} \tag{3.5.1a}$$

$$y_t = Gx_t \tag{3.5.1b}$$

$$d_t = Sx_t \tag{3.5.1c}$$

$$S = HG\left(I - \delta A\right)^{-1} \tag{3.5.1d}$$

where A is a stable matrix, $\delta \in (0,1)$ is a discount factor, and $x_0 \sim \mathcal{N}(\mu, \Sigma)$, where μ, Σ are the mean vector and covariance matrix, respectively, of a stationary distribution for the $\{x_t\}_{t=0}^{\infty}$ process. As usual, w_{t+1} is the time $t+1$ component of an i.i.d. sequence of $\mathcal{N}(0, I)$ random vectors. Here d_t is a vector of decisions that an econometrician observes. Equations (3.5.1c) and (3.5.1d) assert that d_t equals a fixed matrix H times a vector of forecasts of discounted sums of current and future y_t vectors, forecasts that equal the appropriate conditional expectations (please see equation (2.4.17)). To fix ideas with an example, take a univariate autoregression for $\{y_t\}$, namely,

$$y_{t+1} = \alpha + \rho_1 y_t + \rho_2 y_{t-2} + \sigma w_{t+1} \tag{3.5.2}$$

that we can represent with equation (3.5.1a)-(3.5.1b). We can think of d_t as a time t *decision* and $d_t = Sx_t$ as a *decision rule* that tells how an economic agent or a collection of economic agents sets d_t as a function of time t information x_t about future y_{t+j}'s, in particular, so that $d_t = HE_t \sum_{j=0}^{\infty} \delta y_{t+j}$ $\to \delta^j$ where E_t denotes a mathematical expectation conditional on time t information. Lucas (1976) offered three examples that can be fit into this framework. In those examples the decisions d_t were either a choice of a consumption rate or an investment rate or or a labor supply amount (i.e., Lucas's Phillips curve example), and y_t was a stochastic process (e.g., a tax rate or inflation process), forecasts of which influenced time t decisions d_t.[17]

[17] A variety of other examples that we'll encounter later in this book also fit.

Note that by virtue of stationarity, $\begin{bmatrix} d_t \\ x_t \end{bmatrix}$ is a multivariate normal vector with partitioned covariance matrix $\begin{bmatrix} \Sigma_{dd} & S\Sigma_{dx} \\ \Sigma_{xd}S' & \Sigma_{xx} \end{bmatrix}$, where Σ_{xx} satisfies the discrete Lyapunov equation $\Sigma_{xx} = A\Sigma_{xx}A' + CC'$ and $\Sigma_{dx} = S\Sigma_{xx}$. Application of the least squares formula (2.5.3) confirms that S equals a matrix of population regression coefficients of d_t on x_t and that the R^2 for each component of d_t is 1.[18]

Lucas (1976) described and criticized a collection of studies that had used the following faulty logic.

a. A researcher does not understand or care that S satisfies restriction (3.5.1d) but just wants to know the decision rule for d_t .

b. The researcher gathers historical observations on $\{d_t, x_t\}_{t=0}^T$, correctly believes that $d_t = Sx_t$ is a regression equation, and therefore estimates the matrix S of regression coefficients by applying least squares equation-by-equation.

c. Mistakenly as it will turn out, the researcher assumes that the decision rule $d_t = Sx_t$ that he has estimated from historical data will remain invariant under arbitrary hypothetical alterations of the right-hand side variables $\{x_t\}$, including ones that no longer would be described by the historical law of motion (3.5.1a)-(3.5.1b). He believes that he can use that estimated decision rule to predict how systematic variations in $\{x_t\}$ will induce systematic variations in $\{d_t\}$. (see next page)

d. Holding the historically estimated decision rule S fixed, the researcher simulates outcomes $\{d_t\}_{t=T+1}^{\tilde{T}}$ under alternative rules for generating time paths for $\{x_t\}_{t=T+1}^{\tilde{T}}$ [19] and then evaluates the resulting $\{d_t, x_t\}_{t=T+1}^{\tilde{T}}$ outcomes according to a social welfare function that depends on these outcome paths.

e. After learning how the welfare function varies with the hypothetical rule for setting the $\{x_t\}_{t=T+1}^{\tilde{T}}$ sequence, the researcher recommends a continuation $\{x_t\}_{t=T+1}^{\tilde{T}}$ sequence that yields the best social welfare. For example, the researcher holds S fix and uses it as a component of a state-space system

[18] Apply formula (2.5.3) for the matrix of regression coefficients and notice that $\beta = S\Sigma_{xx}\Sigma_{xx}^{-1} = S$ and that the conditional covariance matrix $\hat{\Sigma}_{dd} = 0$.

[19] For example, by using alternative state-dependent "policy rules" to generating them.

- Do you mean "of the distribution"?

Can't he still arbitrarily vary "right hand side" variables under the assumption that the law of motion remains unchanged?

What I mean is the following: given an estimated $\hat{S}$ from $\{d_t, x_t\}_{t=0}^{T}$, <u>under the assumption that S doesn't change</u>, I can make out of sample predictions for d_t for an arbitrary value of x_t. In this sense I can "arbitrarily" vary x_t.

to which he applies a linear-quadratic control theory to be described in chapter 6 to recommend a law of motion for x_t that manifests itself as a new pair of matrices $\tilde{A}, \tilde{G}$ that in the future will replace A, G in the state-space representation (3.5.1), while he holds S fixed at the value estimated from historical data.

f. When decision makers recognize that $\tilde{A}, \tilde{G}$ and not A, G will describe future y_t's, they adjust the decision rule S to become $\tilde{S} = H\tilde{G}(I - \beta\tilde{A})^{-1}$. By failing to understand that, the researcher and policy maker will be disappointed to find that the regression that the researcher estimated on the historical data fails to describe data generated under the regime $\tilde{G}, \tilde{A}$.

Lucas (1976) used this line of reasoning to characterize an influential 1940s-1960s econometric policy evaluation research method that applied optimal control techniques to Keynesian and monetarist econometric models to propose improved macroeconomic policies. Lucas said that the method is flawed because it ignores the key cross-equation restrictions (3.5.1c). Altering the policy-generating matrices from the G, A pair that prevailed historically to new pair $\tilde{G}, \tilde{A}$ that describes a hypothetical policy "experiment" that we want to study amounts to a positing a "regime change" that should be expected systematically to alter how y_t depends on x_t, as described by system (3.5.1). A change in the decision maker's environment G, A alters the moment matrices Σ_{xx}, Σ_{dx} and therefore the population regression matrix of d_t on x_t that will characterize outcomes under the new regime. Thus, structure (3.5.1), and in particular the cross-equation restriction (3.5.1c), imply that S will vary *systematically* with hypothetical changes in $\tilde{G}, \tilde{A}$, belying the assumption that S is invariant to the hypothetical policy interventions being investigated. A concise way to say this is that the joint probability distribution of the $\{d_t, x_t\}$ process and the probability distribution of d_t conditional on x_t process both depend on the (A, C, G) triple of matrices.

3.5.1d?

same

Lucas (1976) summarized his argument as follows:

> *. . . given that the structure of an econometric model consists of optimal decision rules of economic agents, and that optimal decision rules vary systematically with changes in the structure of series relevant to the decision makers, it follows that any change in policies will systematically affect the structure of econometric models.*
>
> — *"Econometric Policy Evaluation: A Critique," 1976*

Equation (3.5.1d) is an effective way of capturing how "optimal decision rules vary systematically with changes in the structure of series relevant to the decision makers." We already encountered an example of such restrictions on optimal decision rules in section 3.3 when we found that the LQ permanent income model of consumption smoothing implies an optimal decision rule for consumption of the form (3.3.11). We shall encounter more examples in chapter 6. As an essential part of a better theory of econometric policy evaluation, Lucas (1976) called for the construction of a rational expectations econometrics that incorporates the kinds of cross-equation restrictions that system (3.5.1) encodes.[20]

3.6. Interpretations and responses to the Lucas Critique

Section 3.5 already mentioned a principal response to the Lucas Critique, namely, rational expectations econometrics, an estimation and interpretation strategy now widely used in applied macroeconomics. In this section, we use a state-space system with coefficients that are functions of a finite state Markov process as a vehicle to describe some related responses that are based on the perspective that changes in government decision rules or "regimes" don't occur in a vacuum but were understood during the historical sample period being modeled econometrically. Thus, this view imagines government decision rules as being governed by statistical processes that are known to the agents inside a rational expectations model. So changes in government decision rules are anticipated and properly responded to in advance by private agents. These ideas adopt and elaborate a perspective of Sargent and Wallace (1976) that regards government policy rules as being determined by a process that leaves econometricians and policy analysts with no "free parameters" to change, disabling them from being able to offer policy advice. Thus, Sargent and Wallace argued that

> *... new rules are not adopted in a vacuum. Something would cause the change- a change in adminstrations, new appointments, and so on. Moreover, if rational agents live in a world in which rules can be and*

[20] See Sargent (1981) and the introductory essay in Lucas and Sargent (1981), a volume containing early contributions to rational expectations econometrics. See Hansen and Sargent (1991a) for further technical contributions.

> *are changed, their behavior should take into account such possibilities and should depend on the process generating the rule changes. But invoking this kind of complete rationality seems to rule out normative economics completely by, in effect, ruling out freedom for the policymaker. For in a model with completely rational expectations, including a rich enough description of policy, it seems impossible to define a sense in which there is any scope for discussing the optimal design of policy rules. That is because the equilibrium values of the endogenous variables already reflect, in the proper way, the parameters describing the authorities prospective subsequent behavior, including the probability that this or that proposal for reforming policy will be adopted.*
>
> *— "Rational Expectations and the Theory of Economic Policy," Sargent and Wallace (1976), p. 181*

This perspective was embraced and extended by Christopher A. Sims (1982) and Sargent (1984). We use tools encountered in chapter to 2 express these ideas.

In describing their work with vector autoregressions with regime shifting coefficients and volatilities, Sims and Zha (2006) link their work to the Lucas Critique:

> *The model with time variation in coefficients in all equations might be expected to fit best if there were policy regime changes, and the nonlinear effects of these changes on private sector dynamics, via changes in private sector forecasting behavior, were important. That this is possible was the main point of Robert E. Lucas (1972). . . . once we recognize that changes in policy must in principle themselves be modeled as stochastic, Lucas's argument can be seen as a claim that a certain sort of nonlinearity is important. Even if the public believes that policy is time-varying and tries to adjust its expectation formation accordingly, its behavior could be well approximated as linear and non-time-varying. As with any use of a linear approximation, it is an empirical matter whether the linear approximation is adequate for a particular sample or counterfactual analysis.*
>
> *— "Were There Regime Switches in U.S. Monetary Policy?" Sims and Zha (2006), p. 59*

Here we'll use chapter 2 tools to represent and study these parts of Sims and Zha's description:

a. How to represent "nonlinearities" in the form of dependencies of state-space coefficients that are functions of a discrete Markov state variable that stands in for a "regime".

b. How agents who understand regime transition probabilities and who recognize the current regime shape their decisions.

c. How a best-fitting fixed-coefficient linear model that ignores regime changes can approximate observed outcomes.

3.6.1. Markov switching linear state space model

We can capture policy regime changes with a linear state space model with coefficients governed by a finite state Markov chain to represent the nonlinearities mentioned by Sims and Zha.[21] Let $s_t \in \{1, 2\}$ be governed by a Markov chain with transition matrix P, where $P_{ij} = \text{Prob}(s_{t+1} = j | s_t = i)$, $\pi_{0,i} = \text{Prob}(s_0 = i)$, and $\pi_{t,i} = \text{Prob}(s_t = i)$. We assume that $\{s_t, x_t, y_t\}_{t=0}^{\infty}$ is governed by the linear state-space system with Markov-switching matrices

$$\begin{aligned} x_{t+1} &= A_{s_t} x_{t+1} + C_{s_t} w_{t+1} \\ y_t &= G_{s_t} x_t \end{aligned} \tag{3.6.1}$$

where x_0 is a random initial vector drawn from density ϕ_0 and $w_{t+1} \sim \mathcal{N}(0, I)$ is an i.i.d. sequence of random vectors. We want to compute the conditional mathematical expectations

$$z_t = E\left(\sum_{t=0}^{\infty} \beta^t y_{t+j}\right) \Big| x_t, s_t$$

and

$$\tilde{z}_t = E\left(\sum_{t=0}^{\infty} \beta^t y_{t+j}\right) \Big| x_t.$$

[21] Hamilton (1989) introduces this type of model and labels it as "non-stationary". If we assume that chain π_0 is a stationary distribution, then the model is actually stationary, a property that we exploit below. Williams and Svensson (2008) and Svensson and Williams (2009) used linear quadratic dynamic programming problems with Markov-switching coefficients to model optimal monetary policy.

We also want to compute an associated time-invariant vector autoregression that is implied by (3.6.1) and that takes the form

$$x_{t+1} = \bar{A}x_t + v_{t+1} \tag{3.6.2}$$

where v_{t+1} is a serially uncorrelated shock with mean zero and contemporaneous covariance matrix $Ev_{t+1}v'_{t+1} = \bar{\Sigma}$. This is the vehicle that Sims and Zha mention when they say of the time-varying system that "its behavior could be well approximated as linear and non-time-varying".

3.6.2. Example

Before computing these objects, let's take an example in the spirit of a rational expectations version of Philip Cagan's (1956) model of hyperinflation. The rate of inflation is the first difference of the logarithm of the price level and the rate of growth of the money supply is the first difference of the logarithm of the money supply. Let p_t be the rate of inflation and let m_t be the rate of growth of the money supply. Assume that

$$p_t = (1-\lambda)\, m_t + \lambda E_t p_{t+1}, \quad \lambda \in (0,1)$$

so that

$$p_t = (1-\lambda)\, E_t \sum_{j=0}^{\infty} \lambda^j m_{t+j},$$

where $E_t(\cdot)$ is a mathematical expectation conditioned on time t information, which will be either x_t or (x_t, s_t), where (x_t, s_t) are defined as follows. We assume that $s_t \in 1, 2$ is governed by a two-state Markov chain P with initial distribution π_0. Setting $x_t = [\,1 \quad m_t \quad m_{t-1}\,]'$ lets us capture the following stochastic process for $\{m_t\}_{t=0}^{\infty}$ with the state-space form (3.6.1):

$$m_{t+1} = \rho_{1,s(t)} m_t + \rho_{2,s(t)} m_{t-1} + c_{s(t)} w_{t+1},$$

where w_{t+1} is an i.i.d. univariate standardized normal random variable.

3.6.3. Conditional distributions of geometric sums

We begin by computing z_t, an expectation conditional on both x_t and s_t. We begin by guessing that

$$z_t = H_{s_t} x_t$$

and substitute this guess into

$$z_t = y_t + \beta E\left[z_{t+1} | x_t, s_t\right]$$

to get

$$H_i = G_i + \beta \sum_j H_j A_i P_{ij},$$

which implies that

$$\begin{aligned} H_1 &= G_1 + \beta\left[H_1 A_1 P_{11} + H_2 A_1 P_{12}\right] \\ H_2 &= G_2 + \beta\left[H_1 A_2 P_{21} + H_2 A_1 P_{22}\right]. \end{aligned}$$

Stacking these equations and solving for the matrices H_1, H_2, we compute

$$\begin{bmatrix} H_1 & H_2 \end{bmatrix} = \begin{bmatrix} G_1 & G_2 \end{bmatrix} \left(I - \beta \begin{bmatrix} A_1 P_{11} & A_2 P_{21} \\ A_1 P_{12} & A_2 P_{22} \end{bmatrix} \right)^{-1}.$$

Next we compute $\tilde{z}_t = E(\sum_{t=0}^{\infty} \beta^t y_{t+j}) | x_t$. We start knowing that the distributions conditional on (x_t, s_t) are multivariate normal:

$$\begin{aligned} x_{t+1} | x_t, s_t &\sim \mathcal{N}\left(A_{s_t} x_t, C_{s_t} C'_{s_t}\right) \\ y_t | x_t, s_t &\sim \mathcal{N}\left(G_{s_t} x_t, 0\right) \\ z_t | x_t, s_t &\sim \mathcal{N}\left(H_{s_t} x_t, 0\right) \end{aligned}$$

Distributions of x_{t+1}, y_t, and z_t conditional on x_t are *mixtures of normals*, with $\pi_{t,i}$ being the mixing probabilities over states i at time t. Defining the matrix averages

$$\begin{aligned} \bar{A}_t &= \pi_{t,1} A_1 + \pi_{t,2} A_2 \\ \bar{G}_t &= \pi_{t,1} G_1 + \pi_{t,2} G_2 \\ \bar{H}_t &= \pi_{t,1} H_1 + \pi_{t,2} H_2, \end{aligned}$$

we can deduce conditional (on x_t) means

$$\begin{aligned} E x_{t+1} | x_t &= \bar{A}_t x_t \\ E y_t | x_t &= \bar{G}_t x_t \\ E z_t | x_t &= \bar{H}_t x_t. \end{aligned}$$

The appearance of time subscripts t in the formulas for $\bar{A}_t, G_t, H_t$ are a consequence of π_t not being a stationary distribution.

Having computed conditional means, let's compute conditional (on x_t) covariances. Since

$$x_{t+1} - \bar{A}x_t = \begin{cases} \left(A_1 - \bar{A}_t\right) x_t + C_1 w_{t+1} & \text{if } s_t = 1 \\ \left(A_2 - \bar{A}_t\right) x_t + C_2 w_{t+1} & \text{if } s_t = 2 \end{cases}$$

it follows that

$$\begin{aligned} E\left(x_{t+1} - \bar{A}_t x_t\right)\left(x_{t+1} - \bar{A}_t x_t\right)' | x_t &= \sum_i \pi_{t,i} \left[\left(A_i - \bar{A}_t\right) x_t x_t' \left(A_i - \bar{A}_t\right)' + C_i C_i'\right] \\ &\equiv \Sigma_t\left(x_t\right) \end{aligned} \tag{3.6.3}$$

where i is present through $\pi_{t,i}$ and x_t is present because it appears on the right side of the first line. Since

$$y_t = \begin{cases} G_1 x_t & \text{if } s_t = 1 \\ G_2 x_t & \text{if } s_t = 2 \end{cases}$$

and

$$y_t - \bar{G}x_t = \begin{cases} \left(G_1 - \bar{G}_t\right) x_t & \text{if } s_t = 1 \\ \left(G_2 - \bar{G}_t\right) x_t & \text{if } s_t = 2 \end{cases}$$

it follows that

$$E\left(y_t - \bar{G}_t x_t\right)\left(y_t - \bar{G}_t x_t\right)' | x_t = \sum_i \pi_{t,i} \left(G_i - \bar{G}_t\right) x_t x_t' \left(G_i - \bar{G}_t\right)'.$$

and similarly that

$$E\left(z_t - \bar{H}_t x_t\right)\left(z_t - \bar{H}_t x_t\right)' | x_t = \sum_i \pi_{t,i} \left(H_i - \bar{H}_t\right) x_t x_t' \left(H_i - \bar{H}_t\right)'.$$

We can compute unconditional means and covariances of the $\{x_t\}_{t=0}^{\infty}$ process as follows. Define the unconditional moments $Ex_t = \mu_t$ and $\Sigma_t = E(x_t - \mu_t)(x_t - \mu_t)'$. Note that

$$x_{t+1} - \mu_{t+1} = \begin{cases} A_1\left(x_t - \mu_t\right) + C_1 w_{t+1} & \text{if } s_t = 1 \\ A_2\left(x_t - \mu_t\right) + C_2 w_{t+1} & \text{if } s_t = 2 \end{cases}$$

It follows that

$$\mu_{t+1} = \bar{A}_t \mu_t$$

and

$$\Sigma_{t+1} = \pi_{t,1}\left(A_1 - \bar{A}_t\right)\Sigma_t\left(A_1 - \bar{A}_t\right)' + \pi_{t,2}\left(A_2 - \bar{A}_t\right)\Sigma_t\left(A_2 - \bar{A}_t\right)' + \left(\bar{C}\bar{C}'\right)_t$$

where $(\bar{C}\bar{C}')_t = \sum_i \pi_{t,i} C_i C_i'$. When $\pi_t \to \bar{\pi}$ as t goes to infinity, iterations on these two recursions converge, respectively, to a stationary mean vector μ and a stationary covariance matrix Σ that satisfy

$$\mu = \bar{A}\mu$$

and

$$\bar{\Sigma} = \bar{\pi}_1\left(A_1 - \bar{A}\right)\bar{\Sigma}\left(A_1 - \bar{A}\right)' + \bar{\pi}_2\left(A_2 - \bar{A}\right)\bar{\Sigma}\left(A_2 - \bar{A}\right)' + \bar{C}\bar{C}'$$

where $\bar{\pi}_i = \lim_{t\to+\infty} \pi_{t,i}$ and $\bar{C}\bar{C}' = \sum_i \bar{\pi}_i C_i C_i'$. A time vector autoregression of the form (3.6.2), namely,

$$x_{t+1} = \bar{A}x_t + v_{t+1}.$$

has $Ev_{t+1}v_{t+1}' = \bar{\Sigma}$.

Quentin R0: Please go over the code in your notebook and reorganize to have all population computations at the front of the notebook. By "population" I mean analytic with our formulas, not the simulations with the least squares regressions. Please put that material at the end of the notebook. Please make the notebook so that Zejin can understand the flow and approve of its readability.

Quentin R1: Please compute and display conveniently for me the objects $C_1C_1', C_2C_2', \bar{\Sigma}, \bar{A}$.

Quentin R2: Please compute "all of our objects" for the above example with two s_t dependent ρ_1, ρ_2 vectors, namely $1.2, -.3$ and $.95, 0$ with a transition matrix $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ using as $\pi_0 = \pi_\infty$ the stationary distribution $\begin{bmatrix} .5 & .5 \end{bmatrix}'$ or with another set of parameter values that I can input easily.

Quentin R3: Please check whether $\bar{H} = \bar{G}(I - \beta\bar{A})^{-1}$ and display outcomes so that Zejin and I can readily spot them.

3.6.4. Time series averages

Consider the conditional covariance matrix

$$\Sigma\left(x_t\right)=\sum_i \pi_{t,i}\left[\left(A_i-\bar{A}\right)x_t x_t'\left(A_i-\bar{A}\right)'+C_i C_i'\right].$$

Assume that the pair (P,π_0) governing the Markov state s_t is stationary. Then the joint process $\{x_t,s_t\}$ process is also stationary, and so is the process $\{x_t\}$. Say that x_t has the unconditional probability distribution $\Phi(x)$. Then a law of large numbers tells us that as $T\to+\infty$

$$T^{-1}\sum_{t=0}^{T}\Sigma\left(x_t\right)\to\int\Sigma\left(x\right)d\Phi\left(x\right)\equiv\Sigma$$

3.7. Conclusions

The examples in this chapter are the foundations of much theoretical and empirical work at the core of modern applied macroeconomics.[22] In the next chapter we study decision problems in which the goal is optimally to manage the evolution of a state vector that can be partially controlled.

[22] For example, see Quah (1990) and Blundell and Preston (1998) for applications of some of the tools of this chapter and of chapter 6 to studying some puzzles associated with a permanent income model.

Exercises

Exercise 3.1 **Consumption**

a. Please use formulas (3.3.19) to verify formulas (3.3.23) and (3.3.25)-(3.3.26) of subsection 3.3.3.

b. Please use formulas (3.2.3) to compute the decision rules in formulas (3.3.23) and (3.3.25) for the following parameter values: $\beta = .95, \sigma_1 = \sigma_2 = 1$.

c. Please use formulas (3.2.3) to compute the decision rules in formulas (3.3.23) and (3.3.25) for the following parameter values: $\beta = .95, \sigma_1 = 2, \sigma_2 = 1$.

d. Please use formula (3.3.21) to confirm formulas (3.3.24) and (3.3.27).

Exercise 3.2 A household has preferences over consumption processes $\{c_t\}_{t=0}^{\infty}$ that are ordered by

$$-.5 \sum_{t=0}^{\infty} \beta^t \left[(c_t - 30)^2 + .000001 b_t^2 \right] \tag{1}$$

where $\beta = .95$. The household chooses a consumption, borrowing plan to maximize (1) subject to the sequence of budget constraints

$$c_t + b_t = \beta b_{t+1} + y_t$$

for $t \geq 0$, where b_0 is an initial condition, β^{-1} is the one-period gross risk-free interest rate, b_t is the household's one-period debt that is due in period t, and y_t is its labor income, which obeys the second-order autoregressive process

$$\left(1 - \rho_1 L - \rho_2 L^2\right) y_{t+1} = (1 - \rho_1 - \rho_2)\, 5 + .05 w_{t+1}$$

where $\rho_1 = 1.3, \rho_2 = -.4$.

a. Define the *state* of the household at t as $x_t = [\,1 \quad b_t \quad y_t \quad y_{t-1}\,]'$ and the *control* as $u_t = (c_t - 30)$. Then express the transition law facing the household in the form (2.4.21). Compute the eigenvalues of A. Compute the zeros of the characteristic polynomial $(1 - \rho_1 z - \rho_2 z^2)$ and compare them with the eigenvalues of A. (*Hint:* To compute the zeros in Matlab, set $a = [\,.4 \quad -1.3 \quad 1\,]$ and call `roots(a)`. The zeros of $(1 - \rho_1 z - \rho_2 z^2)$ equal the *reciprocals* of the eigenvalues of the associated A.)

b. Write a Matlab program that uses the Howard improvement algorithm (2.4.29) to compute the household's optimal decision rule for $u_t = c_t - 30$. Tell how many iterations it takes for this to converge (also tell your convergence criterion).

c. Use the household's optimal decision rule to compute the law of motion for x_t under the optimal decision rule in the form

$$x_{t+1} = (A - BF^*)\, x_t + Cw_{t+1},$$

where $u_t = -F^* x_t$ is the optimal decision rule. Using Matlab, compute the impulse response function of $[\, c_t \quad b_t \,]'$ to w_{t+1}. Compare these with the theoretical expressions (3.3.19).

Exercise 3.3 **Permanent income model again**

Each of two consumers named $i = 1, 2$ has preferences over consumption streams that are ordered by the utility functional

$$E_0 \sum_{t=0}^{\infty} \beta^t u\left(c_t^i\right) \tag{1}$$

where E_t is the mathematical expectation conditioned on the consumer's time t information, c_t^i is time t consumption of consumer i at time t, $u(c)$ is a strictly concave one-period utility function, and $\beta \in (0, 1)$ is a discount factor. The consumer maximizes (1) by choosing a consumption, borrowing plan $\{c_t^i, b_{t+1}^i\}_{t=0}^{\infty}$ subject to the sequence of budget constraints

$$c_t^i + b_t^i = R^{-1} b_{t+1}^i + y_t^i$$

where y_t is an exogenous stationary endowment process, R is a constant gross risk-free interest rate, b_t^i is one-period risk-free debt maturing at t, and $b_0^i = 0$ is a given initial condition. Assume that $R^{-1} = \beta$. We impose the following condition on the consumption, borrowing plan of consumer i:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(b_t^i\right)^2 < +\infty.$$

Assume the quadratic utility function $u(c_t) = -.5(c_t - \gamma)^2$, where $\gamma > 0$ is a bliss level of consumption. Negative consumption rates are allowed.

Let $s_t \in \{0,1\}$ be an i.i.d. process with $\text{Prob}(s_t = 1) = \text{Prob}(s_1 = 0) = .5$. The endowment process of consumer 1 is $y_t^1 = 1 - .5s_t$ and the endowment process of person 2 is $y_t^2 = .5 + .5s_t$. Thus, the two consumers' endowment processes are perfectly negatively correlated i.i.d. processes with means of .75.

a. Find optimal decision rules for consumption for both consumers. Prove that the consumers' optimal decisions imply the following laws of motion for b_t^1, b_t^2:

$$\begin{aligned} b_{t+1}^1 (s_t = 0) &= b_t^1 - .25 \\ b_{t+1}^1 (s_t = 1) &= b_t^1 + .25 \\ b_{t+1}^2 (s_t = 0) &= b_t^2 + .25 \\ b_{t+1}^2 (s_t = 1) &= b_t^2 - .25 \end{aligned}$$

b. Show that for each consumer, c_t^i, b_t^i are co-integrated.

c. Verify that b_{t+1}^i is risk-free in the sense that conditional on information available at time t, it is independent of news arriving at time $t+1$.

d. Verify that with the initial conditions $b_0^1 = b_0^2 = 0$, the following two equalities obtain:

$$\begin{aligned} b_t^1 + b_t^2 &= 0 \quad \forall t \geq 1 \\ c_t^1 + c_t^2 &= 1.5 \quad \forall t \geq 1 \end{aligned}$$

Use these conditions to interpret the decision rules that you have computed as describing a closed pure consumption loans economy in which consumers 1 and 2 borrow and lend with each other and in which the risk-free asset is a one-period IOU from one of the consumers to the other.

e. Define the 'stochastic discount factor of consumer i' as $m_{t+1}^i = \frac{\beta u'(c_{t+1}^i)}{u'(c_t^i)}$. Show that the stochastic discount factors of consumer 1 and 2 are

$$m_{t+1}^1 = \begin{cases} \beta + .25\frac{\beta(1-\beta)}{(\gamma - c_t^1)}, & \text{if } s_{t+1} = 0; \\ \beta - .25\frac{\beta(1-\beta)}{(\gamma - c_t^1)}, & \text{if } s_{t+1} = 1; . \end{cases}$$

$$m_{t+1}^2 = \begin{cases} \beta - .25\frac{\beta(1-\beta)}{(\gamma - c_t^2)}, & \text{if } s_{t+1} = 0; \\ \beta + .25\frac{\beta(1-\beta)}{(\gamma - c_t^2)}, & \text{if } s_{t+1} = 1; . \end{cases}$$

Are the stochastic discount factors of the two consumers equal?

f. Verify that $E_t m_{t+1}^1 = E_t m_{t+1}^2 = \beta$.

Exercise 3.4 **IQ**.

An infinitely lived person's 'true intelligence' $\theta \sim \mathcal{N}(100, 100)$, i.e., mean 100, variance 100. For each date $t \geq 0$, the person takes a 'test' with the outcome being a univariate random variable $y_t = \theta + v_t$, where v_t is an i.i.d. process with distribution $\mathcal{N}(0, 100)$. The person's initial IQ is $IQ_0 = 100$ and at date $t \geq 1$ before the date t test is taken it is $\mathrm{IQ}_t = E\theta|y^{t-1}$, where y^{t-1} is the history of test scores from date 0 until date $t-1$.

a. Give a recursive formula for IQ_t and for $E(\mathrm{IQ}_t - \theta)^2$.

b. Use Matlab to simulate 10 draws of θ and associated paths of y_t, IQ_t for $t = 0, \ldots, 50$.

c. Prove that $\lim_{t\to\infty} E(\mathrm{IQ}_t - \theta)^2 = 0$.

Exercise 3.5 **IQ again**

Latent intelligence is a scalar random variable $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$. An applicant for college takes k tests whose scores are a $k \times 1$ random vector governed by

$$y = \lambda\theta + u \tag{1}$$

where λ is a known $k \times 1$ vector, $u \sim \mathcal{N}(0, \Sigma_u)$, and $Eu\theta = 0$. diagonal, but that is not necessary. The mean and variance parameters $\mu_\theta, \sigma_\theta$ and the covariance matrix Σ_u as well as the vector of "loadings" λ are all known. A testing service summarizes results of the test as a single score $IQ = E[\theta|y]$, the mathematical expectation of the student's θ conditional on his/her test scores y. Please show that IQ is an affine function of y and give formulas for the constant and slope coefficients as functions of the known parameters $\lambda, \mu_\theta, \sigma_\theta, \Sigma_u$.

Exercise 3.6 **Math and verbal IQ's**

An infinitely lived person's 'true intelligence' θ has two components, math ability θ_1 and verbal ability θ_2, where $\theta \sim \mathcal{N}\left(\begin{bmatrix} 100 \\ 100 \end{bmatrix}, \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}\right)$. For each date $t \geq 0$, the person takes a single 'test' with the outcome being a univariate random variable $y_t = G_t\theta + v_t$, where v_t is an i.i.d. process with distribution $\mathcal{N}(0, 50)$ and $G_t = [\,.9 \quad .1\,]$ for $t = 0, 2, 4, \ldots$ and $G_t = [\,.01 \quad .99\,]$ for $t = 1, 3, 5, \ldots$. Here the person takes a math test at t even and a verbal test at

t odd (but you have to know how to read English to survive the math test, and you have to know how to tell time in order to plan your time allocation well for the verbal test). The person's initial IQ vector is $IQ_0 = \begin{bmatrix} 100 \\ 100 \end{bmatrix}$ and at date $t \geq 1$ before the date t test is taken it is $\mathrm{IQ}_t = E\theta|y^{t-1}$, where y^{t-1} is the history of test scores from date 0 until date $t-1$.

a. Give a recursive formula for IQ_t and for $E(\mathrm{IQ}_t - \theta)(\mathrm{IQ}_t - \theta)'$.

b. Use Matlab to simulate 10 draws of θ and associated paths of y_t, IQ_t for $t = 0, \ldots, 50$.

c. Show computationally or analytically that $\lim_{t \to +\infty} E(\mathrm{IQ}_t - \theta)(\mathrm{IQ}_t - \theta)' = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Exercise 3.7 **Invertibility**

A univariate stochastic process y_t has a first-order moving average representation

$$y_t = \epsilon_t - 2\epsilon_{t-1} \tag{1}$$

where $\{\epsilon_t\}$ is an i.i.d. process distributed $\mathcal{N}(0,1)$.

a. Argue that ϵ_t cannot be expressed as as linear combination of $y_{t-j}, j \geq 0$ where the sum of the squares of the weights is finite. This means that ϵ_t is not in the space spanned by square summable linear combinations of the infinite history y^t.

b. Write equation (1) as a state space system, indicating the matrices A, C, G.

c. Using the matlab program `kfilter.m` to compute an innovations representation for $\{y_t\}$. Verify that the innovations representation for y_t can be represented as

$$y_t = a_t - .5a_{t-1} \tag{2}$$

where $a_t = y_t - E[y_t|y^{t-1}]$ is a serially uncorrelated process. Compute the variance of a_t. Is it larger or smaller than the variance of ϵ_t?

d. Find an autoregressive representation for y_t of the form

$$y_t = \sum_{j=1}^{\infty} A_j y_{t-j} + a_t \tag{3}$$

where $Ea_t y_{t-j} = 0$ for $j \geq 1$. (*Hint:* either use formula (2) or else remember formula (2.9.3).)

e. Is y_t Markov? Is $[\, y_t \quad y_{t-1} \,]'$ Markov? Is $[\, y_t \quad y_{t-1} \quad \cdots \quad y_{t-10} \,]'$ Markov?

f. Extra credit. Verify that ϵ_t *can* be expressed as a square summable linear combination of $y_{t+j}, j \geq 1$.

Exercise 3.8 **Muth and Kalman again**

Consider the linear state space system

$$\begin{aligned} x_{t+1} &= \rho x_t + c w_{t+1} \\ y_t &= x_t + v_t, \end{aligned} \tag{1}$$

where $\rho \in [0, 1)$ and c are scalars, $\{w_t\}_{t=0}^{\infty}$ is an i.i.d. scalar process with $w_t \sim \mathcal{N}(0, 1)$, $\{v_t\}_{t=0}^{\infty}$ is an i.i.d. process of measurement errors with $E v_t w_{s+1} = 0$ for all $t \geq 0$ and $s \geq 0$ with $v_t \sim \mathcal{N}(0, r)$ with $r > 0$, and x_0 is a random initial value distributed as $x_0 \sim \mathcal{N}(0, \frac{c^2}{1-\rho^2})$. (Note: Muth (1960) considered the limiting case in which $\rho \uparrow 1$.)

a. Is the univariate process $\{x_t\}_{t=0}^{\infty}$ Markov?

b. Please compute the stationary distribution of the stochastic process $\{x_t\}$ and compare it to the initial distribution assumed here.

c. Is the univariate process $\{y_t\}_{t=0}^{\infty}$ Markov?

d. Find an *innovations representation* for $\{y_t\}_{t=-\infty}^{\infty}$ of the time-invariant form

$$\begin{aligned} \hat{x}_{t+1} &= \rho \hat{x}_t + k a_t \\ y_t &= \hat{x}_t + a_t, \end{aligned} \tag{2}$$

where k is a regression coefficient, $\hat{x}_t = E x_t | y^{t-1}$, $a_t = y_t - E[y_t | y^{t-1}]$, a_t is by construction serially uncorrelated and homoskedastic with $E[a_t | y^{t-1}] = 0$ and $E a_t^2 = \omega$, and y^{t-1} is the semi-infinite history $y_t, y_{t-1}, \ldots$. Please fill in the blanks in this sentence: "k is the regression coefficient of _____ on _____." Please give formulas for k and ω.

e. Please verify the following formulas:

$$\begin{aligned} \hat{x}_{t+1} &= k \sum_{j=0}^{\infty} (\rho - k)^j \, y_{t-j} \\ y_t &= k \sum_{j=0}^{\infty} (\rho - k)^j \, y_{t-j-1} + a_t, \end{aligned}$$

where a_t is orthogonal to y_{t-j-1} for $j \geq 1$ so that the second equation is an *autoregressive representation* or *autoregression* for y_t. Please verify that $E[y_{t+1}|y^t] = \hat{x}_{t+1}$.

f. Where $s^t = [s_t, s_{t-1}, \ldots, s_0]$ for a scalar process $\{s_t\}_{t=-\infty}^{\infty}$, please verify the following formulas:

$$\begin{aligned} x_t &= c \sum_{j=0}^{\infty} \rho^j w_{t-j} \\ y_t &= c \sum_{j=0}^{\infty} \rho^j w_{t-j} + v_t \\ E\left[y_t | w^{t-1}, v^{t-1}\right] &= c \sum_{j=1}^{\infty} \rho^j w_{t-j} \\ E\left(y_t - \left[y_t | w^{t-1}, v^{t-1}\right]\right)^2 &= c^2 + r. \end{aligned}$$

Please verify that the prediction error variance $E(y_t - [y_t|w^{t-1}, v^{t-1}])^2$ is less than the prediction error variance Ea_t^2. Please explain why.

g. Where L is the lag operator presented in appendix B of chapter 2, please verify the following formula:

$$(\dagger) \qquad a_t = \frac{1}{1 - (\rho - k) L} \left[c w_t + (1 - \rho L) v_t\right].$$

Please interpret this formula as showing how the contemporaneous scalar shock a_t reflects "old news" in the form of past w_t's and past v_t's. Please interpret the discrepancy in variances of one-step ahead prediction errors that you computed in part **f** in light of this formula.

h. Show that while $\frac{1}{1-(\rho-k)L}[cw_t]$ and $\frac{1}{1-(\rho-k)L}[(1-\rho L)v_t]$ are both serially correlated processes, their sum a_t is not. Please interpret this peculiar situation.

Exercise 3.9 **From shocks to innovations, I**

Consider again the setting described in exercise *3.8*. Let $\{y_t\}_{t=-\infty}^{\infty}$ be the covariance stationary scalar stochastic process appearing in exercise *3.8*. We begin by introducing a covariance generating function defined as the z-transform of the autocovariance sequence $\{c_y(\tau)\}_{\tau=-\infty}^{\infty}$, where $c_y(\tau) = Ey_t y_{t-\tau}$. The covariance generating function is

$$S_y(z) = \sum_{j=\infty}^{\infty} c_y(j) z^j.$$

Using the original state-space representation for y_t, it can be verified that

$$\begin{aligned} S_y(z) &= r + \left(\frac{c}{1-\rho z}\right)\left(\frac{c}{1-\rho z^{-1}}\right) \\ &= \frac{c^2 + r(1-\rho z)\left(1-\rho z^{-1}\right)}{(1-\rho z)(1-\rho z)} \\ &= \frac{c^2 + r\left(1+\rho^2\right) - r\rho z - r\rho z^{-1}}{(1-\rho z)\left(1-\rho z^{-1}\right)}. \end{aligned}$$

(The spectral density defined in section 2.10 is the covariance generating function obtained by setting $z = e^{-i\omega}, \omega \in [-\pi, \pi]$.) Using the innovations representation for $\{y_t\}$ instead of the original state-space representation cast in terms of the hidden state x_t, it can be verified that the covariance generating function also equals

$$S_y(z) = \frac{\omega(1-(\rho-k)z)\left(1-(\rho-k)z^{-1}\right)}{(1-\rho z)\left(1-\rho z^{-1}\right)}.$$

The denominators in both representations of $S_y(z)$ are equal and the numerators in both representations must also be equal. Therefore, it must be true that

$$(1) \qquad c^2 + r\left(1+\rho^2\right) - r\rho z - r\rho z^{-1} = \omega(1-(\rho-k)z)\left(1-(\rho-k)z^{-1}\right).$$

The left side is a quadratic polynomial in z. The right side factors this polynomial. The zeros of the quadratic polynomial in z evidently form a reciprocal pair $(\rho-k), (\rho-k)^{-1}$.[23] Equation (1) as an instance of a more general *spectral factorization identity* (see Hansen and Sargent (2013, p. 164).) Also, please note the steady state Kalman filter generates a Kalman gain k that is a key to achieving the factorization.

Now take representation (†) for a_t from exercise *3.8* and use it to compute the covariance generating function of the innovation process $\{a_t\}$:

$$\begin{aligned} S_a(z) &= \frac{c^2}{(1-(\rho-k)z)(1-(\rho-k)z^{-1})} + \frac{r(1-\rho z)\left(1-\rho z^{-1}\right)}{(1-(\rho-k)z)(1-(\rho-k)z^{-1})} \\ &= \frac{c^2 + r\left(1+\rho^2\right) - r\rho z - r\rho z^{-1}}{(1-(\rho-k)z)(1-(\rho-k)z^{-1})}. \end{aligned}$$

[23] Such a reciprocal pairs property characterizes large classes of linear filtering problems and linear optimal control problems. We shall encounter more examples in chapter 6.

It follows from the preceding equation and factorization identity (1) that

$$S_a(z) = \omega,$$

which implies that $c_a(j) = 0$ for $j \neq 0$ and $c_a(0) = \omega$. This confirms that $\{a_t\}$ is a serially uncorrelated process with variance ω.

a. Please verify the preceding calculations.

b. Please plot the spectral density $S_a(e^{-i\omega})$ against $\omega \in [0, \pi]$ and compare it with those of the scalar processes depicted in Figures 2.10.2, 2.10.3, 2.10.1, and 2.10.4 from section 2.10.

Exercise 3.10 **From shocks to innovations, II**

Consider a state space system and the associated time-invariant innovations representation, which we repeat here for convenience. The original system is

$$\begin{aligned} x_{t+1} &= Ax_t + Cw_{t+1} \\ y_t &= Gx_t + v_t \end{aligned} \tag{1}$$

where $\{w_{t+1}\}_{t=0}^{\infty}$ is an i.i.d. sequence with $w_{t+1} \sim \mathcal{N}(0, I)$ for all t and $\{v_t\}_{t=0}^{\infty}$ is an i.i.d. sequence with $v_t \sim \mathcal{N}(0, R)$ for all t; and $Ew_t'v_s = 0$ for all t and s. The associated time-invariant "innovations representation" state-space system is

$$\begin{aligned} \hat{x}_{t+1} &= A\hat{x}_t + Ka_t \\ y_t &= G\hat{x}_t + a_t. \end{aligned} \tag{2}$$

Here Cw_{t+1} are the shocks hitting the hidden state x_{t+1} conditional on x_t, v_t are shocks or measurements hitting y_t conditional on Gx_t, and $a_t = y_t - E[y_t|y^{t-1}]$ are shocks hitting y_t given the infinite history $y^{t-1} = [y_{t-1}, y_{t-2}, \ldots]$. The shock a_t is i.i.d. with distribution $a_t \sim \mathcal{N}(0, \Omega)$.

a. Please provide formulas for K and Ω.

b. Please deduce from the above pair of state-space systems a single state-space system that maps the shocks w_{t+1}, v_{t+1} into the innovation a_{t+1} and verify that it takes the form:

$$
\begin{aligned}
\begin{bmatrix} x_{t+1} \\ \hat{x}_t \\ v_{t+1} \end{bmatrix} &= \begin{bmatrix} A & 0 & 0 \\ KG & (A-KG) & K \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ \hat{x}_t \\ v_t \end{bmatrix} + \begin{bmatrix} C & 0 \\ 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} w_{t+1} \\ v_{t+1} \end{bmatrix} \\
a_t &= \begin{bmatrix} G & -G & I \end{bmatrix} \begin{bmatrix} x_t \\ \hat{x}_t \\ v_t \end{bmatrix}
\end{aligned} \tag{3}
$$

c. Please use formulas from section 2.4 to compute moving average representations and autocovariograms for the process $\{a_t\}$. Is $\{a_t\}$ a serially correlated process?

d. Please use formulas from part **c.** to deduce impulse response functions of a_t to w_t and of a_t to v_t so that $a_t = \sum_{j=0}^{\infty} g_j w_{t-j} + \sum_{j=0}^{\infty} h_j v_{t-j}$. Please represent the autocovariance function of $\{a_t\}$ as the sum of the autocovariance functions of $\sum_{j=0}^{\infty} g_j w_{t-j}$ and $\sum_{j=0}^{\infty} h_j v_{t-j}$ and verify that $E a_t a_{t-j}' = 0$ for $j \neq 0$. (Feel free to write a Matlab or Python program to perform these calculations and verifications.)

Exercise 3.11 **A monopolist, learning, and ergodicity**

A monopolist produces a quantity Q_t of a single good in every period $t \geq 0$ at zero cost. At the beginning of each period $t \geq 0$, before output price p_t is observed, the monopolist sets quantity Q_t to maximize

$$E_{t-1} p_t Q_t \tag{1}$$

where p_t satisfies the linear inverse demand curve

$$p_t = a - bQ_t + \sigma_p \epsilon_t \tag{2}$$

where $b > 0$ is a constant known to the firm, ϵ_t is an i.i.d. scalar with distribution $\epsilon_t \sim \mathcal{N}(0,1)$, and the constant in the inverse demand curve a is a scalar random variable unknown to the firm and whose unconditional distribution is $a \sim \mathcal{N}(\mu_a, \sigma_a^2)$, where $\mu_a > 0$ is large relative to $\sigma_a > 0$. Assume that the random variable a is independent of ϵ_t for all t. Before the firm chooses Q_0, it knows the unconditional distribution of a, but not the realized value of a. For each $t \geq 0$, the firm wants to estimate a because it wants to make a good decision about output Q_t. At the end of each period t, when it must set Q_{t+1}, the firm observes p_t and also of course knows the value of Q_t that it had set. In (1), for $t \geq 1$, $E_{t-1}(\cdot)$ denotes the mathematical expectation conditional on the history of signals $p_s, q_s, s = 0, \ldots, t-1$; for $t = 0$, $E_{t-1}(\cdot)$ denotes the expectation conditioned on no previous observations of p_t, Q_t.

a. What is the optimal setting for Q_0? For each date $t \geq 0$, determine the firm's optimal setting for Q_t as a function of the information p^{t-1}, Q^{t-1} that the firm has when it sets Q_t.

b. Under the firm's optimal policy, is the pair (p_t, Q_t) Markov?

c. 'Finding the state is an art.' Find a recursive representation of the firm's optimal policy for setting Q_t for $t \geq 0$. Interpret the state variables that you propose.

d. Under the firm's optimal rule for setting Q_t, does the random variable $E_{t-1} p_t$ converge to a constant as $t \to +\infty$? If so, prove that it does and find the limiting value. If not, tell why it does not converge.

e. Now suppose that instead of maximizing (1) each period, there is a single infinitely lived monopolist who once and for all before time 0 chooses a plan for

an entire sequence $\{Q_t\}_{t=0}^{\infty}$, where the Q_t component has to be a measurable function of (p^{t-1}, q^{t-1}), and where the monopolist's objective is to maximize

$$E_{-1} \sum_{t=0}^{\infty} \beta^t p_t Q_t \tag{3}$$

where $\beta \in (0,1)$ and E_{-1} denotes the mathematical expectation conditioned on the null history. Get as far as you can in deriving the monopolist's optimal sequence of decision rules.